
Lecture 21: Vector Fields

Background & Motivation

We now begin the final chapter of the course: vector calculus. Until now, our functions have been scalar-valued— $f(x, y)$ returns a number. A vector field $\mathbf{F}(x, y)$ returns a vector at each point: think wind velocity, magnetic force, or fluid flow. Over the next six lectures, we'll integrate vector fields along curves (line integrals), across surfaces (flux integrals), and connect everything through the great theorems of Green, Stokes, and Gauss. These theorems unify all of calculus into a single principle: the integral of a derivative over a region equals the integral of the function over the boundary.

1. Warm-Up: Gradient Review

Find ∇f for $f(x, y, z) = x^2y + yz^3$.

1. $f_x =$ _____
2. $f_y =$ _____
3. $f_z =$ _____
4. $\nabla f =$ _____

– The gradient assigns a *vector* to each point – this is an example of a **vector field**.

2. Vector Field Definition

Definition 1: Vector Field

A **vector field** on $\mathbb{R}^2$ is a function $\mathbf{F}(x, y) = P(x, y)\hat{\mathbf{i}} + Q(x, y)\hat{\mathbf{j}}$.

A **vector field** on $\mathbb{R}^3$ is $\mathbf{F}(x, y, z) = P\hat{\mathbf{i}} + Q\hat{\mathbf{j}} + R\hat{\mathbf{k}}$.

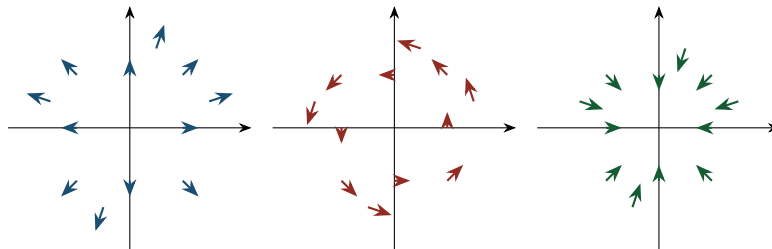
- At each point, $\mathbf{F}$ assigns a vector (direction and magnitude)
- Examples: wind velocity, gravitational force, electric field, fluid flow

3. How to Draw a Vector Field

Procedure for Sketching $\mathbf{F}(x, y)$

1. Pick a grid of sample points (e.g., $(\pm 1, 0)$, $(0, \pm 1)$, $(\pm 1, \pm 1)$, etc.)
 2. At each point (a, b) : evaluate $\mathbf{F}(a, b) = \langle P(a, b), Q(a, b) \rangle$
 3. Draw an arrow **starting at** (a, b) , pointing in the direction $\mathbf{F}(a, b)$
 4. Arrow length $\propto \|\mathbf{F}(a, b)\|$ (longer arrow = stronger field)
- **Key:** the arrow's *tail* is at the sample point, not its midpoint
 - **Normalization:** $\hat{\mathbf{F}} = \mathbf{F}/\|\mathbf{F}\|$ makes all arrows length 1 (shows direction only)

Common Patterns



Source: $\mathbf{F} = \langle x, y \rangle$ **Rotation:** $\mathbf{F} = \langle -y, x \rangle$ **Sink:** $\mathbf{F} = \langle -x, -y \rangle$

- **Source:** vectors point away from origin – fluid is being “created”
- **Sink:** vectors point toward origin – fluid is being “absorbed”
- **Rotation:** vectors swirl around a point – “whirlpool” pattern

4. Examples

Example 1: Evaluating and Sketching a Rotation Field

Let $\mathbf{F}(x, y) = \langle -y, x \rangle$. Evaluate at several points and describe the pattern.

Example 2: Evaluating a Source Field

Let $\mathbf{F}(x, y) = \langle x, y \rangle$. Evaluate at $(1, 0)$, $(0, 2)$, $(-1, 1)$.

5. Conservative Vector Fields

Definition 2: Conservative Field

$\mathbf{F}$ is **conservative** if there exists a scalar function f (the **potential function**) such that:

$$\mathbf{F} = \nabla f$$

- Work done by a conservative field is _____
- In physics: conservative $\iff$ energy is conserved (e.g., gravity, electrostatics)

Theorem 1: Test for Conservative Fields (2D)

$\mathbf{F} = \langle P, Q \rangle$ is conservative on a simply connected domain if and only if:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

3D version: $\mathbf{F} = \langle P, Q, R \rangle$ is conservative iff $\nabla \times \mathbf{F} = \mathbf{0}$:

$$P_y = Q_x, \quad P_z = R_x, \quad Q_z = R_y$$

Caution: The domain must be **simply connected** (no holes) for the test to be sufficient.

Simply Connected Domains

- **Simply connected** – every closed curve can be shrunk to a point without leaving the domain
- $\mathbb{R}^2$ and $\mathbb{R}^3$: simply connected ✓
- $\mathbb{R}^2 \setminus \{(0, 0)\}$: **not** simply connected (has a “hole”)

6. Finding Potential Functions

Algorithm for Finding f (2D)

Given $\mathbf{F} = \langle P, Q \rangle$ that passes the conservative test:

1. **Integrate:** $f = \int P \, dx$ (treat y as constant) $\implies f = (\dots) + g(y)$
2. **Differentiate:** $f_y = (\dots) + g'(y)$. Set equal to Q and solve for $g'(y)$.
3. **Integrate** $g'(y)$ to find $g(y)$. Combine.

For 3D: same idea but with $g(y, z)$, then $h(z)$.

Example 3: Testing and Finding a Potential Function

Is $\mathbf{F} = \langle 2xy + y^2 \cos x, x^2 + 2y \sin x \rangle$ conservative? If so, find f .

Example 4: Non-Conservative Field: The Hole Counterexample

Show that $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ passes $P_y = Q_x$ but is NOT conservative.

Practice Problems

Problem 1. Evaluate $\mathbf{F}(x, y) = \langle x^2, xy \rangle$ at $(1, 2)$, $(2, 1)$, $(-1, 3)$. For each, state where the arrow starts and what direction it points.

Problem 2. Determine if $\mathbf{F} = \langle ye^{xy}, xe^{xy} \rangle$ is conservative. If so, find f .

Problem 3. Show that $\mathbf{F} = \langle x^2 - y^2, 2xy \rangle$ is NOT conservative.

Problem 4. Find the potential function for $\mathbf{F} = \langle 2xz + y, x + z^2, x^2 + 2yz \rangle$.

Problem 5. Let $\mathbf{F} = \langle y \cos(xy) + 1, x \cos(xy) + 2z, 2y + 3z^2 \rangle$.

- (a) Verify $\mathbf{F}$ is conservative (check all three conditions).
- (b) Find the potential function.